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Free-roam location-based virtual reality entertainment: An affordance analysis

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ABSTRACT

Free-roam location-based virtual reality entertainment (free-roam LBEVR) is reshaping the visitor economy by combining spacious physical venues and synchronized roamable VR systems to produce immersive experiences. Yet little attention has been paid to this intersection between entertainment and tourism. Drawing on qualitative exploration combining participatory observation, user-generated content retrieval, and semi-structured interviews, this paper identified four core affordances, including story-ability, gamifiability, roam-ability, and physical interactivity that supply the free-roam LBEVR experiences. These affordances jointly enable somatic presence, embodied sense-making, and heightened subjectivity, while producing an ineffability paradox whereby rich, embodied experiences resist verbal representations. Theoretically, the present study conceptualizes free-roam LBEVRs as site-centric experiential infrastructures with potential functioning as tourist attractions, and introduces somatic presence to extend the understanding of phygital experiences. Practically, our findings guide designers and destination managers to align material space and digital content, prioritize embodied cues, and innovate marketing strategies to convey ineffable experiences.

1. Introduction

Free-roam location-based virtual reality entertainment (free-roam LBEVR) allows visitors to physically navigate expansive digital environments (Fig. 1). For example, the *Horizon of Khufu* depicts the 4500-year-old Great Pyramid of Giza (Lao, 2025). Drawing over 100,000 visitors in its first ten months in Shanghai (Jingan Shanghai, 2023), this attraction signals the growing market appeal of immersive, movement-based digital experiences, a trend also reflected in the broader expansion of the location-based entertainment (LBE) sector, whose global market is projected to reach USD 25.90 billion by 2030, with an estimated compound annual growth rate of 28.5% (Grand View Research, 2024, pp. 2025–2030).

Importantly, free-roam LBEVR represents a novel intersection between VR applications and LBE. Unlike traditional LBE rooted in physical rides or at-home VR confined by sedentary boundaries, free-roam LBEVR leverages spacious rooms and wearable gear, including headsets, trackers, and haptic vests, to offer embodied, mediated experiences aligning physical movements with virtual content. While some destinations, such as Luoyang, China, have successfully deployed free-roam LBEVR as standalone tourist attractions (Ministry of Cultural and Tourism China, 2023), academic attention to its unique operational and experiential dynamics remains limited. Specifically, two gaps exist in the current literature.

First, existing frameworks derived from sedentary VR or location-based AR are insufficient to explain the high degree of embodied participation in free-roam LBEVR. Previous theorization primarily examined visual- and auditory-dominated applications where physical movement is restricted (Lacka, 2018; Lv et al., 2025; Zhang et al., 2024). In contrast, free-roam LBEVR requires visitors to navigate mapped physical spaces, enhancing both embodiment and real-time social co-presence (Zheng et al., 2025). The integration of body movement and virtual representation requires a new conceptual lens beyond traditional VR theories. Second, VR in tourism is predominantly studied as an online, marketing-oriented tool (Calisto & Sarkar, 2024; Skard et al., 2021), leaving its role as site-centric recreational infrastructure under-explored. Explicating how reality display venues are engineered to deliver immersive outcomes may improve the understanding of in-destination consumable entertainment.

We adopted an affordance perspective in analysis. Affordance

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denotes action possibilities arising from actor-environment relations (Gibson, 2014). It is useful to unveil how environments, such as landscapes, built settings, and mobile technologies, shape tourist behaviors (Chen & Wu, 2021; Tomej & Xiang, 2020). As a relational and emergent construct, affordance bridges macro structures and micro agency (Evans et al., 2017), functioning as a robust analytic tool for understanding the experience-environment nexus. Taken together, two research questions were formulated to guide the present investigation:

1. What are free-roam LBEVR's affordances?
2. What are the experiential consequences arising from these affordances?

A three-phase, multimodal qualitative approach was employed. Phase 1 involved participatory observation to document material and technological elements in free-roam LBEVR. Phase 2 harvested naturally occurring user-generated content (UGC) to capture visitor experience representations. Phase 3 comprised semi-structured interviews with both visitors and managers to obtain deeper, situated perspectives. Data were analyzed through an interpretive lens to identify how the designed spaces afford experience-making.

The present study is expected to make twofold contributions. First and theoretically, this study reconceptualizes free-roam LBEVR as a tourist-attraction-like experiential infrastructure orchestrated with physical space and digital content. It also extends the understanding of LBE and tourist on-site entertainment, particularly in how extended reality technologies afford innovative experiences with elements including somatic immersion and embodiment. Second and pragmatically, our findings illustrate the experience-environment nexus and deconstruct how technologies enable or disable warranted consumer experiences. This offers actionable guidance for project designers and destination managers in the deployment of free-roam LBEVR.

2. Literature review

2.1. Entertainment, LBE, and tourism

Free-roam LBEVR represents a recent innovation within the entertainment industry, which focuses on providing designed activities that produce pleasurable experiences and satisfying leisure time (Pearce, 2008b; Vogel, 2014). Entertainment is a well-established contributor to the hospitality and travel sectors (Lin et al., 2025; Luo & Lam, 2017; Pearce, 2008a, 2008b), serving as an inspiration for travel (Florido-Benítez, 2023), a primary destination attractor that builds competitiveness (Liang & Li, 2021), and a crucial on-site consumable that stimulates expenditure and extends visitation length (Milman, 2008; Williams, 2020).

LBE represents a specific subset of entertainment experienced at physical venues outside the home (Ford, 2022). While the broader entertainment industry includes mass media and location-free distribution, LBE is rooted in traditional collective amusements, such as amphitheaters and festivals, which rely on specific out-of-home spaces for shared experiences. Consequently, LBE emphasizes venue-based staging and experiential qualities that are strictly contingent upon physical presence. However, the traditional LBE model faces inherent constraints. Venues are often capital-intensive and spatially rigid, which limits scalability and restricts both geographic reach and content evolution (Luo & Lam, 2017). Digital innovation provides unique solutions. Digital content creates immersive worlds without the need for heavy construction (Wang et al., 2024). This digital turn establishes VR as a viable mechanism for constructing and evolving destination attractions.

2.2. Free-roam LBEVR: A new possibility for tourist attraction

Free-roam LBEVR represents a significant evolution in digitalized entertainment. Unlike conventional VR formats (e.g., standalone VR, tethered VR, cave VR, or phone-based VR) and other location-tracking-based applications (e.g., location-based AR games), free-roam LBEVR allows visitors to navigate spacious mapped environments through wireless headsets and peripheral equipment such as trackers, haptic devices, or hand controllers. This configuration facilitates precise avatar-body synchronization across the physical and virtual spaces, thereby shifting VR from a primarily visual or screen-based medium to a bodily enacted, site-centric attraction. Despite its growing adoption and experiential uniqueness, research on the operational and experiential dynamics of free-roam LBEVR remains insufficient.



Fig. 1. An example of free-roam LBEVR

Some useful insights can be drawn from existing literature on location-based augmented reality (AR) in tourism (see Appendix A). Location-based AR utilizes GPS and geographical points of interest to create movement-dependent gameplay, such as *Pokémon Go* (Mihale-Wilson et al., 2021) or destination-specific interpretation tools (Zhang et al., 2024). Moreover, a futuristic LBE trend is location-based mixed reality (MR) applications. MR blends virtual and physical content to interact in spatially consistent ways, and like location-based AR, ties content to geo-locations (Santoso et al., 2022; Tongpaeng et al., 2024). While both AR and MR hold potential as tourist attractions, their deployment is often hindered by high costs and case-specific content requirements (Zhang et al., 2024). In contrast, free-roam LBEVR leverages more controlled environments and standardized service procedures, making it more viable for mass deployment as in-destination consumable entertainment.

Additionally, free-roam LBEVR is distinct from traditional VR applications typically used for pre-visit marketing or post-visit savoring (Lee et al., 2021; Li et al., 2024). While most tourism research examines VR as a medium for armchair travel, free-roam LBEVR functions as on-site experiential infrastructure. Its venue-dependent nature requires physical visitation, positioning it as a standalone attraction or a primary component of site-centric entertainment rather than a promotional tool (Buhalis et al., 2022, 2023). It also outstands current sedentary-VR-based on-site facilities in incorporating walkable spaces for bodily participation (Wang et al., 2024). This shift from a marketing-oriented medium to a site-centric attraction necessitates a re-evaluation of how VR is blended into the tourism system.

Last, free-roam LBEVR introduces experience-making improvements. Traditional VR provides experiences dominated by visual and auditory stimuli, frequently failing to instantiate somatic and materially anchored relations (Chen et al., 2025; Wang et al., 2026). While previous studies emphasize virtual presence and narrative consumption as primary drivers of immersion (Hu et al., 2021; Wei et al., 2019), free-roam LBEVR prioritizes bodily synchronization. Consequently, existing frameworks derived from sedentary VR are insufficient for explaining the embodied dynamics of free-roam LBEVR. Thus, examining the experience-environment nexus in free-roam LBEVR can add nuances to the understanding of VR experience construction.

2.3. An Affordance perspective

Affordance theory provides a robust framework for understanding the experience-environment nexus. It denotes the action possibilities that environments offer to agents (Gibson, 1977, 2014). Rather than a simple one-to-one property, affordance is a layered, relational, and multi-directional construct (Evans et al., 2017; Ronzhyn et al., 2022). Functioning as a meso-level explanatory layer between the environment and the actor, affordances emerge when environmental features match actor capacities, allowing a setting to support multiple potential actions.

In tourism, the affordance perspective serves as a useful tool for understanding and designing experiences because it reveals how environments guide, enable, or constrain visitor practices (McCabe, 2024; Stinson et al., 2022). The affordance-oriented focus then underscores the contextualization of action possibilities. For example, Tomej and Xiang (2020) deconstructed tourism product affordances for visually impaired travelers into categories such as follow-ability, traverse-ability, and touch-ability. By linking specific material features to supported practices, affordance-based analysis helps identify connections or discrepancies between physical forms and the goals of experience engineering.

Affordance theory has been widely applied to digitally enriched experiences. Studies of tourism live streaming and short-video platforms, for example, investigated how various affordances such as visibility, meta-voicing, searchability, interactivity, etc., produce immersion and stimulate viewer engagement and consumption intentions (Liu et al., 2024; Zhang et al., 2025). Some have applied the affordance perspective to unpack immersive-experience-making with advanced reality displays (Chen & Kalay, 2008; Yun et al., 2023). These applications demonstrate that affordance thinking is useful for experience engineering in the digitalized context that the present investigation concerns.

Importantly, how free-roam LBEVR affords visitors' experiences is worthy- but under-explored. As these venues proliferate, they offer experiences comprising interrelated layers of physical contact, immersive audiovisual rendering, and real-time social interaction. How these dimensions are hybridized and simultaneously afforded through embodied infrastructure and representational devices requires further explication. Adopting an affordance perspective can capture how embodied action possibilities emerge relationally between visitors and the spatial-technological environment, providing a comprehensive understanding of free-roam LBEVR experience. While existing work on VR, XR, and AR affordances provides a foundation, it often falls short of theorizing the multisensory, large-scale dynamics of free-roam spaces (Lee et al., 2025; Liu et al., 2024; Zhang et al., 2025). Consequently, this investigation seeks to articulate the specific affordances that enable visitor experiences in this context, providing new insights into the evolution of digitalized tourist attractions.

3. Methodology

A multi-method qualitative design was employed. We began with participatory observations to document material staging and technological affordances, followed by the collection of UGC to capture visitor experience representations. Finally, semi-structured interviews were conducted with both suppliers and consumers for further insights. The study focuses on the Chinese context. Driven by proactive governmental support and a consumer base open to digitally mediated consumption, China presents itself as an informative context where a sufficient base of creators, managers, and visitors is accessible for empirical study.

3.1. Data collection

Participatory observation was conducted to obtain an emic perspective of the embodied and multisensory nature of free-roam LBEVR. Relying solely on etic materials risked misrepresenting the lived experience. Our participatory engagement explored how material elements enable specific affordances and their subsequent experiential consequences. We participated in 14 free-roam LBEVR programs (Table 1). Before each session, a pre-session template was completed to

Table 1
Free-roam LBEVR where participatory observation was conducted.

EN name or translation	Producer Company	Original Chinese title
The Horizon of Khufu	Excurio, France	消失的法老
Eternal Notre-Dame	Excurio, France	永恒的巴黎圣母院
Titanic: Echoes from the Past	Eclipso, France	泰坦尼克号:遗落的船奇
Winds of Luoyang	**X-Meta, China**	**风起洛阳**
Kunlun Maze	**X-Meta, China**	**昆仑迷宫**
Qin: Awakening	Yuan Tuo Jing, China	秦潮觉醒
Following the Long March	Sai Ai Ning, China	跟着走·长征
Glory of the Silk Road	Sheng Shun, China	丝路荣耀
Panda's World	**OpenWorld, China**	**熊猫的世界**
Galactic Rhapsody	**OpenWorld, China**	**启程**
Exploring the Nation of *Haihun*	**OpenWorld, China**	**探秘海昏侯国**
Nobody	Film Infinity, China	浪浪山小妖怪·妖你同行
Journey to the West	**Sky Limit Entertainment, China**	**西游降魔**
Antarctic Odyssey	**Utour digital, China**	**身临极境·南极起航**
Echoes of Jingdezhen	**Utour digital, China**	**何以景德镇**

Note: Free-roam LBEVRs with bold text were brought into cross-case examination (see section 3.2.).

record the venue, equipment, and group composition. As recording was prohibited during the sessions, the researchers relied on post-participation notes. Within 60 min of exiting each session, fieldnotes and memos were produced to link observed moments to potential affordances, ensuring that the emic observations remained rigorous and amenable to systematic analysis.

Following the fieldwork, multimodal UGC was retrieved to triangulate observation data and further capture visitors' naturalistic representations. Textual data were harvested from Little Red Book and Weibo, China's influential social media platforms (Chen et al., 2021), while audiovisual records were collected from Douyin and Bilibili (Xu et al., 2021). Search keywords included Chinese terms for free-roam LBEVR and the names of popular programs identified via sales rankings on Dianping.com. These criteria were negotiated and co-determined by the research team (Table 2). The collection period spanned from April to June 2025. To ensure data quality, company-generated posts and entries containing fewer than five words were excluded. The final dataset comprised 917 entries, including 879 textual posts (totaling 62,350 characters) and 38 videos.

Last, semi-structured interviews were conducted with visitors, designers, and managers. This allowed the study to incorporate both demand- and the supply-side reflections. Participants were recruited through purposive and snowball sampling, starting with four visitors identified via social platforms and expanding through their networks. After interviewing nine visitors, two additional participants were approached to confirm that no new themes emerged, indicating saturation. Furthermore, eight industry professionals were interviewed, including entrepreneurs, content creators, and tech developers. All sessions were audio-recorded, supplemented by handwritten notes, and transcribed verbatim for analysis. Appendix B provides detailed profiles of the participants.

3.2. Data analysis

Thematic analysis was conducted following Braun and Clarke (2006) six-step procedure. We incorporated both deductive and inductive coding approaches. The analysis started with preliminary readings, reliving emic experiences, and note-taking, followed by iterative readings to ensure familiarity with the data. Initial codes were generated deductively by labelling meaningful segments of the data. These codes were then collated into potential themes reflecting broader patterns. This was followed by careful review and refinement to ensure consistency and distinctiveness. Subsequently, the identified themes were refined. Finally, the analysis was synthesized into a coherent report, illustrated with representative data extracts and linked to the research objectives. The coding scheme was negotiated through an iterative consensus-building process. Two authors conducted the initial coding, which was then audited by the third author. Interpretive discrepancies were discussed throughout the process. Revisitation and re-evaluation were then conducted to achieve shared understanding.

Table 2
Keywords for UGC search.

Search category	English translation	Original Chinese keyword
Expressions signifying free-roam LBEVR	VR big space/free-roam VR	VR大空间
	Big space VR/free-roam VR	大空间VR
	VR arcade	VR游乐场
	VR exhibition	VR展览
Popular free-roam LBEVR	Winds of Luoyang	风起洛阳
	X-meta	机遇时空X-meta
	The Horizon of Khufu	消失的法老
	Love Between Fairy and Devil	苍兰决
	The Origins of Lives	生命起源
	The Disappeared Oracle Bone Inscriptions	消失的甲骨文
	Qin: Awakening	秦潮觉醒

Although all three datasets were brought to the same analysis, the operational emphases were strategically different. Participatory observation served as the coding baseline, providing an emic, on-site foundation that documented the overall experiences (i.e., experiential outcomes) and helped reveal the affordances (i.e., the four main affordances). The UGC data were integrated to triangulate these observations, providing etic evidence of how affordances were perceived and articulated. Finally, interviews provided reflective depth, validating the coding categories, further probing the experiential outcomes, and incorporating supply-side intents. All three types of data appear in the findings report and the coding scheme (i.e., Appendix C).

To strengthen the validity of our findings, we conducted a cross-case comparison between LBEVR projects where participatory observation and supply-side interviews were both conducted (i.e., a total of 8 free-roam LBEVR, see Table 1). Variance in affordance and experiential outcomes was seen among these cases, and cross-case comparisons identified the key mechanism to achieve greater values. The data were collected and analyzed in Chinese, whereas the final report was composed in English. AI-assisted translation was employed and carefully reviewed by the authors to guarantee precision.

4. Findings

4.1. An Affordance breakdown of free-roam LBEVR

We identified four sets of affordances (i.e., story-ability, gamifiability, roam-ability, and physical interactivity) of free-roam LBEVR. Each is composed of sub-category affordances explicating its enactment in practice (as shown in Appendix C).

4.1.1. Story-ability

Story-ability refers to the capability of free-roam LBEVR to scaffold narrative coherence while making the storyworld legible, enterable, and enactable through narratability, roleplay-ability, and liminal transition. Narratability is the most central, which utilizes gated plot beats (e.g., triggers, scripted set pieces, and diegetic artifacts) and multimodal signposting (e.g., voiceovers, on-scene props, and environmental changes) to introduce and reinforce the narrative. For example, in *Titanic: Echo from the Past,* the plot is utilized to organize the visitors' movement. Visitors begin by investigating the wreckage, discover fragments of the past through artifacts and visual clues, and gradually return to the key moments aboard the ship, thereby constructing a coherent narrative progression.

Closely linked to narratability is roleplay-ability, the capability for participants to embody or perform specific narrative roles through identity cues, scripted behaviors, and in-role decision-making. By adopting a first-person perspective, players anchor narrative responsibilities across co-located groups, which enhances individual comprehension of the story structure. In *Titanic*, the user's role as a diver provides a functional and narrative lens through which the virtual environment is navigated and understood. Together, these affordances transform visitors from passive observers into active participants within the story's framework.

Finally, liminal transition facilitates a meaningful separation from the physical world through ritual participation. This is achieved by treating the headset as a separation signifier, where wearing the devices becomes a ceremonial boundary-crossing into the narrative. The orchestration of the threshold and pre-play space is critical to this affordance. For example, *Winds of Luoyang* utilizes themed architecture, in-character greeters, and costume pieces to transform the lobby into a liminal zone (Fig. 2). These embodied cues synchronize group timing and establish a common affective orientation, reducing narrative ambiguity and ensuring the intended story remains legible.

4.1.2. Gamifiability

Gamifiability denotes the capability of free-roam LBEVR to structure



Fig. 2. Antique decoration of the entrance to free-roam LBEVR the *Winds of Luoyang*.

participation as a game-like activity. It is a fourfold design-centric property including goal orientation, feedback, reward, and sociality. Game-like activities are initiated through goal orientation, where objectives are embedded either explicitly via mission prompts or implicitly through emergent goals driven by player curiosity. A structured goal architecture encourages participants to navigate the virtual spatiality of free-roam systems, engage with the narrative, and interact with both peers and non-player characters. Complementing this, feedback mechanisms close the loop between action and understanding. Instant, multimodal feedback, such as haptic sensations, spatialized auditory, visual state changes, and reactive non-player characters (NPCs), confirms the consequences of user actions, allowing players to refine their strategies and maintain a state of flow.

Finally, reward systems and socialization determine whether these goals and feedback loops translate into sustained engagement. Rewards in free-roam LBEVR range from narrative payoffs, such as rankings on a leaderboard, to instrumental incentives like digital collectibles. Socialization remains intrinsic to the format. The multi-user nature of free-roam LBEVR creates social connectivity and provides social proof of participation, further anchoring the user within the gamified environment.

4.1.3. Roam-ability

A primary distinction between free-roam LBEVR and earlier VR formats is roam-ability, defined as the capability to support embodied exploration by enabling movement through physical and virtual spaces and aligning physical locomotion with visual and auditory representations. It is a property of navigational and representational design underpinned by visibility, hear-ability, and traverse-ability.

With the support of location-based systems, simultaneous localization and mapping technologies, and wireless VR, free-roam LBEVR can plant headset-based digital representations into a spatial coordinate system anchored within the physical space. This allows visitors' physical roaming to be translated into synchronized and meaningful movements embodied in their avatars, thus supporting traverse-ability. Such synchronization enhances visibility, as virtual settings become visually legible, spatially coherent, and easy to explore, while also strengthening hear-ability by providing auditory cues that support spatial orientation, environmental awareness, and movement through the virtual environment. Subsequently, both visual and auditory representations become spatially authentic and immersive. Roaming is experienced as embodied exploration rather than simply movement across disconnected scenes. Our emic notes stress that it is the seamless integration between the headset-provided, virtual environment and the physical environment that supplies roam-ability:

After I put on the headset, I enter the world of free-roam LBEVR. I can see human-shaped avatars of myself and peer participants. My movements—hand gestures, arm waiving, walking, leaning, bending, etc. are all synchronized with my avatar. Specifically, the visual and audio representations construct a virtual site or destination, and the rest of my body can feel vicarious presence because of synchronized movements. This synchronization enables a sense of traversing, exploration, and wandering. (fieldnotes by authors)

4.1.4. Physical interactivity

Physical interactivity refers to the capability of free-roam LBEVR to supply actions through tangible, multisensory, and motional interplay. While roam-ability (see Section 4.1.3.) facilitates navigation and exploration, physical interactivity emphasizes manipulative agency. Physical interactivity varies significantly across projects because it necessitates specialized physical engineering to align material settings with digital representations. That said, physical interactivity concerns how the material environment modulates participants' bodily engagement, including tangibility (i.e., encountering material presence aligning with visual representations), atmospheric simulation (i.e., the stimulations of winds, temperature, and scent), and kinaesthetic coupling (i.e., movement-related bodily coupling). It enables participants to shift focus from device-level features to embodied interaction, which supplies a more analytically coherent explanation of how virtual scenes are physically grounded.

One primary method is the physical actualization of virtual representations, where physical settings are designed to be tangible, enabling visitors to hold, manipulate, and physically engage with virtual objects through props or haptic proxies. For instance, in *Winds of Luoyang*, material props such as bamboo raft railings are positioned in the physical space to match virtual counterparts. Additionally, physical

interactivity is operated atmospherically by deploying sensory inputs beyond visual and audio categories. Cues such as airflow, temperature, scent, humidity, and other ambient sensations are integrated to refine realism, deliver environmental meaning, and give bodily responses, as the following transcripts from a vlog made by a gaming influencer indicate:

As the scene unfolded, I entered an ice-and-snow environment. A blast of cold wind carrying the scent of ice suddenly reached my body. In the lava-filled cavern during the boss fight, waves of heat rolled underfoot. At that moment, I was extremely excited. The whole journey combines the virtual with the real. When I entered the lift platform, I could firmly feel the handrail in my hands, and the lift's movements were audible beneath my feet. When I mounted the dragon's back and began to fly, I could feel my body sway with the motion. When I descended the platform and inched along the cliff face, I could reach out and touch an actual rock wall. (videographic UGC transcripts about Mission Ice Loong)

Additionally, kinaesthetic coupling refers to movement that is registered through bodily sway or posture. As the quote demonstrates, riding brought a clear side-to-side swaying sensation, which allows the virtual environment to be encountered as a physical entity acting on, surrounding, and interacting with the body.

4.2. Experiential outcomes

The identified affordances converge to construct immersive experiences within free-roam LBEVR. The data frequently emphasize experiential novelty, noting a stronger engagement nature than that of traditional entertainment formats. Essentially, these experiences emerged through heightened bodily engagement as affordances interact to produce phygitality, a seamless integration of physical and digital worlds designed to create a unified experience (Batat, 2024). In the analysis, we explicated four experiential outcomes. Somatic presence marks the most immediate consequence as the virtual world is increasingly experienced as real. This phygital grounding further extends to sense-making and reclaimed subjectivity since meaning interpretation and negotiation are encouraged through embodied participation. Last, ineffability appears by the end of the journey, experiencing free-roam LBEVR at a representational level, where the embodied intensity surpasses what can be readily articulated in verbal descriptions.

4.2.1. Somatic presence

When the boat crossed the river, there was real-world smoke, mist, and even wind. The seats vibrated in sync with the visuals, simulating the ripples of flowing water or the mist on the surface, perfectly synchronizing my physical perceptions with my visual experience. In addition, my hands would unconsciously reach for the railings to maintain balance when riding in a carriage, and guess what, I could touch the railings or seats in the real world, exactly where my eyes saw them. This consistency between touch and sight instantly convinced me that the ancient world was real, enhancing the feeling of being there. (UGC relevant to Winds of Luoyang)

The above description captures bodily confirmation, the key to somatic presence. Somatic presence concerns a state in which the virtual world is perceived as bodily real through synchronized movements and multisensory feedback. In this regard, the body serves as the primary medium through which the existence of the virtual environment is verified. Sensing, exploring, and confirming the environment through bodily contact makes somatic presence. Different from the presence concept referring to a general subjective state of *being there*, somatic presence is specialized in the sensorimotor synchronization between one's physical body and their digital avatar. In other words, somatic presence is an environment-referential category of presence emerging from corporal validation of virtual representations.

Additionally, somatic presence is distinct from embodiment. Embodiment, especially in the digital-related context, typically refers to the sense of body ownership and agency over a virtual avatar (Chen et al., 2023). In contrast, somatic presence does not merely concern the feeling of having a body. It concerns the validation of the virtual world's reality through the body's interaction with physical, multisensory cues, which often functions as a gate towards reality-testing. Consequently, somatic presence is not just about *being* in a virtual space, but about *believing* in the reality of that space because the physical environment provides the necessary sensory evidence to substantiate the digital representations.

Somatic presence is therefore central to free-roam LBEVR experience, and roam-ability and physical interactivity are the key pillar affordances sustaining this experience. The virtual world is treated by the body as real, not merely due to visual conviction or avatar embodiment, but through ongoing tactile, kinaesthetic, and physiological confirmation during action. This shifts visitors from observing a virtual scene to engaging with it as a bodily consequential environment. Once the virtual environment is somatically verified, visitors' movements, choices, and reactions become experientially meaningful, providing the foundation for subsequent embodied sense-making and the restoration of visitor subjectivity.

4.2.2. Sense-making through embodied metaphors

Somatic presence, with the possibility to enact stories and narratives (i.e., story-ability), further allows meaning-making within free-roam LBEVR. Specifically, sense-making through embodied metaphors, a cognitive process where abstract concepts are derived from recurring sensory-motor experiences (Barsalou, 2008; Wang et al., 2019), is prominent. This sense-making arises from story-infused metaphorical and emotional significance obtained through physical postures, orientations, and tensions.

In free-roam LBEVR, it unfolds in two phases. First, bodily posture and spatial orientation are elevated into abstract meaning. For instance, a physical upward gaze may be mapped to higher-order concepts of power, sublimity, or awe:

Upon entering the main hall, I instinctively craned my neck to look up at the statue of the pharaoh. I felt as small as dust when standing at its feet, as if I were just a tiny ant. At that moment, the plot faded, leaving only the feeling of "looking up" that's so strong and the sense of an unquestionable power within it. (interview data related to The Horizon of Khufu)

Second, bodily tension and instinctive reaction anchor deeper embodied metaphors. For example, by guiding bodily actions through exaggerated scale and height, free-roam LBEVR makes power and sublimity experientially salient through associating height and physical peril. An interviewee experiencing Kunlun Maze highlighted the tension between somatic response and cognitive awareness when walking along a virtual precipice:

There was a section where we walked along the edge of a cliff. I knew there was flat ground beneath my feet, and my friends also told me it was very safe. But when I look down and see the abyss below, my legs involuntarily tighten, and my toes grip the ground tightly, and even dare not go forward. At that moment, my brain tells me, "It's ok, and the scene is fake", but my body won't trust it. And the thought "if I fall, I'm done for" always lingers in my mind. (interview data related to Kunlun Maze)

This demonstrates how bodily actions can override rational knowledge, anchoring the narrative in instinctive physical responses, and the biological metaphor of falling is death. Overall, free-roam LBEVR mobilizes somatic experience to metaphorically reconstruct narrative understanding and emotional resonance. Such experiences are arguably more profound than those offered by traditional VR applications, as the narrative meaning is reinforced through kinesthetic memory rather than passive observation.

4.2.3. Reclaiming subjectivity

Further built on somatic presence and sense-making, the participatory nature of free-roam LBEVR encourages subjectivity restoration. Unlike traditional sightseeing or performance, free-roam LBEVR enables visitors to act as protagonists within the narrative with gamified elements. Action possibilities to fight, hide, and explore foster a belief in

their role as active characters, generating a strong sense of control and self-efficacy, as evidenced by participant reflections like "*I can do*", "*I am involved in*", and "*I have the power to drive the story forward*".

This heightened protagonist agency facilitates a more meaningful escape. Free-roam LBEVR utilizes embodied participation to anchor the visitor in a new mode of existence. By physically enacting the narrative, visitors do not merely observe history but inhabit it. Participants described genuinely inhabiting ancient worlds or embodying historical figures, such as Red Army soldiers, by physically experiencing the hardships of climbing snow-capped mountains. This temporary dissolution of temporal, spatial, and self-boundaries imbues the experience with profound meaning, moving beyond mere entertainment to offer a surreal sense of being.

Furthermore, the intensity of this embodied focus fosters a state of mindfulness that can be psychologically rewarding. The demand for total physical and cognitive engagement forces visitors to detach from external anxieties and become fully present in the moment. As one interviewee reflected: *This project is the most surprising part of this trip. Not only are the beautiful scenes, but we could also learn about history, experience flying and shooting, and the feeling of being in control and physical participation are incredibly strong. I felt totally immersed. It is the most unforgettable part.* This implies that free-roam LBEVR has a standalone restorative effect. The integration of physical agency and virtual sensation creates a therapeutic respite from daily life.

4.2.4. The paradox of ineffability

Beyond the on-site experience, in reflecting and representing the experiences on social media, the intensification of embodied experience may cause a paradox of ineffability. Visitors consistently struggled to translate phygital sensations into verbal descriptions. These recurring patterns of verbal inadequacy, such as the use of condensed evaluations, the absence of narrative elaboration, and the appeal to direct experience, served as empirical evidence of ineffability, rather than seeking linguistic richness as its indicator.

The reasons behind ineffability are twofold. First, on-site recording is often prohibited or technically impractical in free-roam LBEVR once the VR headset is put on and locked in. This forces visitors to rely on post-visit recollection. Thus, much videographic UGC about free-roam LBEVR is made in monologue instead of rich on-site visual representations (Fig. 3). Second, because the experience is primarily kinesthetic and sensory rather than a linear and textual narrative, it resists being easily expanded into a detailed, coherent story. When visitors attempt to share these experiences on social media, they often find that language fails to capture the intensity of the moment. Consequently, they resort to highly condensed evaluative phrases, such as "it was so impressive" or "you just have to experience it for yourself". Additional descriptions are often about the overall narratives and fragments of sensory, bodily encounters in the format of a monologue, sometimes accompanied by videographic montage originally produced by the free-roam LBEVR producers available online.

The brief verbal expression shown in Fig. 3 reflects a cognitive gap between embodied memory and verbal expression. The experience is felt through the body, and when that intensity cannot be translated into words, the only remaining truth is the recommendation of firsthand participation. This ineffability paradox presents a fundamental dilemma for the marketing of free-roam LBEVR. The core value of the experience lies in its exclusivity to the participant's body, making it inherently resistant to second-hand storytelling. As Wang, founder of OpenWorld, who specializes in free-roam LBEVR venue engineering, shared:

We are confident that free-roam LBEVR can have a bright future. The film market is corrupted because of cloud storage, high-definition video representation, etc. But before VR headsets become extremely affordable, people need to get out of their homes to experience free-roam LBEVR.

4.3. Phygital alignment for better experiential actualization

Analyzing affordances-experience nexuses naturally concerns how affordances are designed and implemented for experience actualization. We compared 8 free-roam LBEVR where participatory observation and supply-side interviews were both conducted (see Table 1). The comparison indicates that phygital alignment serves as a critical consideration.

Phygital alignment is a design principle underscoring the coordination between bodily movement, material cues, and virtual representations. That said, the key mission is to cohere the different affordances for joint experience-creation. It stresses on the rhythm of progression, ensuring that visitors' physical movements are meaningfully integrated into the physical and virtual environments. Importantly, it is distinct from physical interactivity. Physical interactivity focuses on offering sensory availabilities, whilst phygital alignment concerns the orchestration of these elements. Without this alignment, interactivity remains a collection of gimmicks. In this sense, phygital alignment acts as a multi-sensory script that guides the design of free-roam LBEVR.



Fig. 3. Screenshots from videographic UGC about *The Horizon of Khufu*.

Li, the lead designer at Utour Digital, repeatedly attributes their success to phygital alignment: "*It is about finding a better way that can work within the limits of existing technology and hardware to truly realize the feelings, the immersion, and the thrill that visitors are looking for. We spent efforts on aligning the virtual and the real. … Take the Jingdezhen project, for example. There is a real technical breakthrough there. When you go up the stairs, you get that vertical physical sensation of upstairing. You do not have that in other projects. It is a purely visual solution to physical sensation. It feels right because leaping happens one step at a time. Our technical director named it 3D Spatial Walking because everything used to be flat. Even on a flying carpet, you are just moving horizontally. This is a vertical visual shift that makes your body follow what your eyes are seeing.*"

Overall, phygital alignment is achieved through three levels of orchestration. The first is sensorimotor integration, whereby virtual representations are designed to anticipate and respond to human movement. This anticipation subtly nudges visitors' bodies into specific postures or movements that reinforce the narrative. Second, the transformation of elements in the physical environment into semantic anchors for the virtual world is critical for enhanced experiences. We labelled it as semantic mapping. For example, if a visitor is walking along stairs, the physical space must be designed to provide the exact upward or downward texture that the brain expects, either physical stairs or representational technologies, as the Utour Digital example. Last, a harmonious rhythm of experiences requires spatiotemporal synchrony. Providing bodily and sensory feedback is not enough. The response must occur at the exact moment of contact. Phygital alignment demands that the physical and virtual inputs arrive at the brain as a single, coherent stream of information. A failed synchronization reminds visitors of technological mediation and pulls them out of immersion.

5. Discussions and conclusions

By adopting an affordance lens, we identified four core categories, including story-ability, gamifiability, roam-ability, and physical interactivity. They are materially instantiated through the spatial and technological features of free-roam LBEVR. These affordances provide action possibilities that become experientially consequential through phygital alignment design. In visitor experience processes, story-ability and gamifiability orient visitors' actions, roam-ability spatializes those actions, and physical interactivity corporeally verifies them. These coordinated affordances generate somatic presence, which further enables sense-making through embodied metaphors and allows visitors to reclaim subjectivity through agency. Ultimately, these dynamics produce a paradox of ineffability, where the depth of embodied experience resists verbal description. Fig. 4 presents the overall mechanisms identified in this investigation.

5.1. Theoretical implications

First, this investigation advances the theoretical understanding of VR-based attractions by shifting the discourse from visual immersion to somatic presence as a bodily grounded mode of experience. While previous tourism scholarship has largely conceptualized VR as a visual substitute for physical travel or a marketing preview tool (Calisto & Sarkar, 2024; Skard et al., 2021), our findings position free-roam LBEVR as a standalone, site-centric destination resource requiring physical participation. This challenges the assumption that digital attractions are inherently virtual replicas of physical sites (Buhalis et al., 2023). Instead, digital attractions like free-roam LBEVR represent a new means of attraction-making where the site is not defined by static geography but by the orchestrated synchronization of physical space and digital content.

This evolving approach to tourist attractions necessitates a re-evaluation of key concepts such as authenticity. Free-roam LBEVR, which often operates in empty warehouses devoid of historical materiality, challenges the objectivist authenticity and aligns more closely with constructivist and existential perspectives (Wang, 1999). Here, experiential truth is derived not from the toured object, but from the

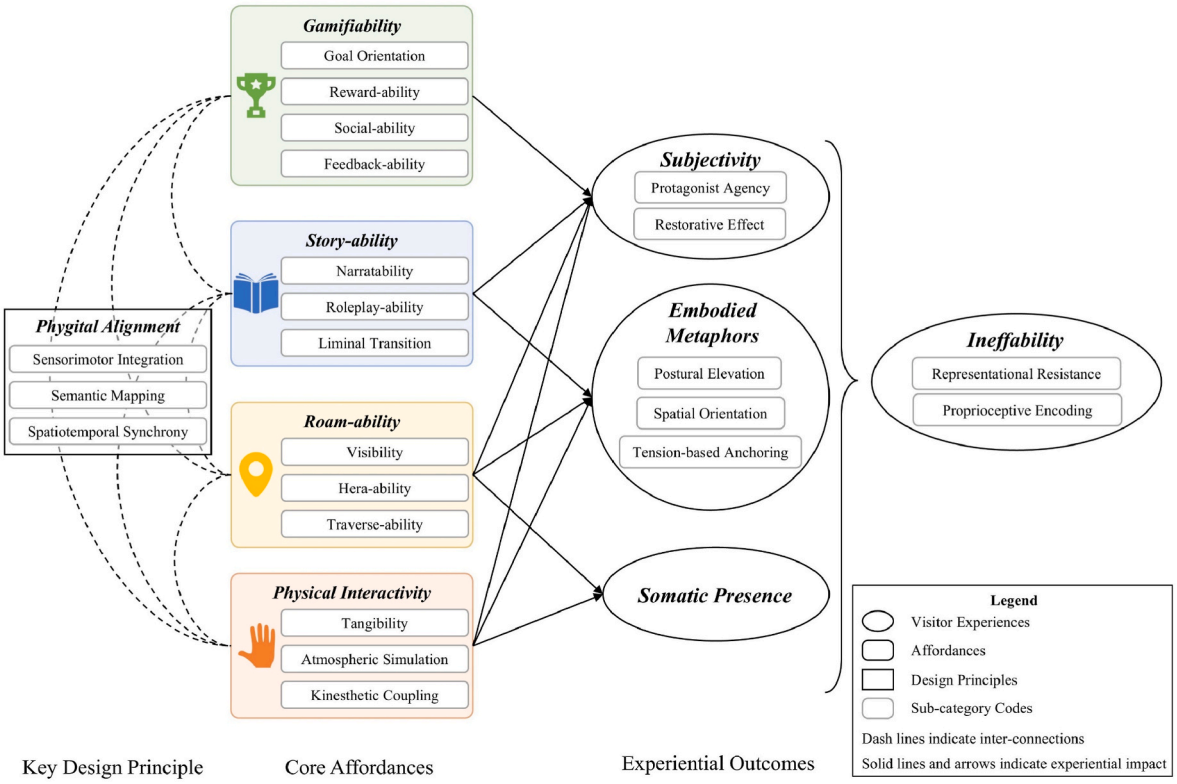


Fig. 4. Design and experience mechanisms centering around free-roam LBVER affordances.

activated state of the subject, a form of hot authentication characterized by intense emotionality and participatory engagement (Cohen & Cohen, 2012). In this context, attraction validity is verified by somatic presence rather than historical provenance. The experience feels authentic because the visitor's body responds to the phygital environment as a tangible reality. Consequently, authenticity in digital visitation becomes less dependent on material history and more on the sincerity of the embodied performance. An interesting paradox is that the more virtual the experience, the more indispensable that the presence of body becomes, which deserves future investigation and theoretical development.

Second, our findings illuminate the evolving convergence of the entertainment and tourism sectors. While entertainment has long driven tourism (Luo & Lam, 2017), free-roam LBEVR acts as a technological bridge that resolves traditional frictions between the two. Traditional LBE venues are typically capital-intensive and spatially rigid (Lee et al., 2020; Liang & Li, 2021), whereas digital entertainment is scalable but often lacks the destination status required for tourism. Free-roam LBEVR hybridizes these attributes by utilizing scalable digital content while maintaining the physical venue requirements of LBE. This fusion creates a consumable, fluid destination that validates entertainment as a primary tourism infrastructure rather than a supplementary activity. Future destination competitiveness may increasingly depend on the capability to host these flexible, phygital spaces.

Third, this work advances the understanding of experience design through an affordance perspective (Luo et al., 2026; Miao et al., 2025; Zheng et al., 2025). By deconstructing the delivery of such experience into specific action possibilities, we move beyond the vague descriptor of phygitality or immersion in the free-roam LBEVR context (Lao, 2025). Importantly, the core affordances and the key design principle of phygital alignment are fundamentally rooted in the technological and spatial configurations of the LBEVR system, making them generalizable across study cases. Certain experiential dimensions, such as collective engagement, may be culturally amplified by the specific case and the Chinese context, but the overall structure of the experiential outcomes (i.e., somatic presence leading to embodied metaphor and subjectivity) is transferable in general.

Specifically, successful experience design in phygital settings like free-roam LBEVR is neither about maximizing graphical fidelity, nor about physical site engineering, but about phygital alignment. That is ensuring that the physical environment affords the same action or sensation that the virtual narrative suggests. This provides a granular theoretical vocabulary for analyzing how built environments and digital overlays collaborate to guide tourist behaviors and shape their perceptions.

Finally, this research contributes to the understanding of VR experiences and phygital visitation. We introduced the concept of somatic presence to describe a state where the physical body validates virtual reality, contrasting with the disembodied observer status common in sedentary VR. It is a niche category of presence state and stresses on a distinct environmental focus compared to the concept of embodiment. Additionally, identifying how somatic presence contributes to embodied metaphors (Barsalou, 2008; Wang et al., 2019), and physical agency for subjectivity and well-being (Tussyadiah et al., 2017) is another contribution to unveiling the experiential benefits of digital technology. The potential to reclaim subjectivity and foster mindfulness and flow suggests that the restorative potential of tourism can be engineered through high-agency phygital environments. This offers a theoretical basis for digital respite in contemporary experience design (Wang et al., 2026).

5.2. Pragmatic implications

This study offers strategic insights for destination managers seeking to integrate free-roam LBEVR as a solution for creating attractions without the need for heavy construction. By facilitating the visitation of lost heritage (e.g., Notre-Dame), inaccessible landscapes (e.g., The Horizon of Khufu), or immovable sites (e.g., the Great Wall), this technology effectively decouples tourism products from geographical and physical limitations. Free-roam LBEVR also introduces action possibilities that were previously restricted, such as gamified interactions within heritage sites. Consequently, operators should view these venues not merely as arcades, but as dynamic cultural institutions capable of refreshing content through software updates. Furthermore, their compact footprint relative to traditional theme parks makes them ideal for urban regeneration or shopping mall revitalization, transforming underused indoor spaces into high-yield tourist destinations.

Additionally, the above experiential outcomes clarify how free-roam LBEVR can shift from technological novelty to a more sustainable tourism attraction. Somatic presence makes the experience difficult to substitute with ordinary screen-based entertainment, as visitors value not only what they see but also what their bodies can verify and enact. On the other hand, embodied sense-making and reclaimed subjectivity help transform spectacular scenes into experiences that are memorable, communicable, and personally involving. Consequently, through connecting bodily sensations with cultural, historical, or narrative meanings, and by allowing visitors to feel that they are actively shaping the experience, free-roam LBEVR can move beyond momentary novelty toward more durable visitor value. These implications suggest that rather than treat free-roam LBEVR as a visual-content business, managers should consider it as an embodied attraction system in which bodily confirmation, meaning construction, and agency jointly shape visitor value.

For practitioners, this research informs the design, management, and marketing of phygital experiences. Designers must prioritize physical interactivity, as bodily confirmation is a critical driver of overall phygitality. Investment in physical props remains essential, as these peripheral cues trigger the somatic presence that defines the core experience. Marketing practitioners must also address the paradox of ineffability. Because these phygital experiences are difficult to describe verbally and on-site recording is often restricted, traditional advertising may prove insufficient. Marketers should therefore innovate methods for UGC creation. For instance, post-visit visualizations can juxtapose a user's physical movements with their virtual perspective, thereby visualizing agency and facilitating social sharing. Additionally, experience-based testimonials can complement visual aesthetics to better communicate the unique, embodied nature of free-roam LBEVR.

5.3. Limitations and future research

This investigation has several limitations. First, the data collection was restricted to the Chinese market. While China is a leader in free-roam LBEVR adoption, cultural variations in technology acceptance and social interaction norms may limit the generalizability of these findings to Western contexts. Second, the study focused exclusively on free-roam LBEVR. As the boundaries between virtual, augmented, mixed, and extended realities continue to blur, future research should examine how this affordance framework applies to other emerging immersive formats. Third, the present study discovered the experiential outcomes of free-roam LBEVR. Further research may extend the proposed framework by investigating downstream behavioral consequences (e.g., satisfaction, revisit intention) and long-term effects (e.g., restorative potential). Fourth, this study does not account for individual heterogeneity in digital familiarity or prior VR exposure, which may modulate how visitors engage with free-roam LBEVR. Future studies are encouraged for further moderation examination. Fifth, textual and verbal data offer limited access to reveal the ineffability of the free-roam LBEVR experience. Future studies may use physiological measures, embodied interviews, or video elicitation to capture these bodily aspects more directly.

CRediT authorship contribution statement

Xuanjin Wu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Xiaomeng Du:** Writing – review & editing, Investigation, Formal analysis, Data curation, Conceptualization. **Ding Xu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declarations of AI usage

GenAI (ChatGPT) was used for language proofing.

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Declarations of interest

We are not aware of any potential conflict of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhtm.2026.101501.

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